

Capstone 2 – Executive Summary

Project Information

Project Title: K'Nex't Level Engineering

Design Team: Turner Marks, Reyna Rodriguez, Dante Scalfi, Jordan Spencer

Design Advisor: Stuart Jacobson

Advisor Contact Email: s.jacobson@northeastern.edu

Abstract

K'Nex roller coaster sets are widely used in K-8th grade classrooms to teach energy and motion concepts. However, gravity-driven model coasters exhibit uneven kinematic scaling when geometrically scaled to real-world coasters. Since gravity cannot be proportionally scaled, the resulting motion deviates from full-scale coaster behavior, leaving the potential for misconceptions in concepts such as energy and motion. This project aims to increase the model's educational value by developing a motorized coaster cart (See Figure 1 below) capable of controlling the train's motion that better visualizes real-world kinematics.



Figure 1: Cart with Motor and Electronics (No Shell)

1. The Need for the Project

Hands-on, experiential learning is used in K–12 STEM education; construction kits like K’Nex are commonly a hands-on device used to teach physics concepts [1][2]. However, gravity-driven model coasters do not clearly represent the behavior of full-scale amusement park roller coasters, creating potential misconceptions in concepts such as motion and energy [3]. This discrepancy stems from scaling limitations, as gravity cannot be adjusted alongside geometric dimensions, resulting in mismatched kinematic behavior [3]. Addressing this gap ensures that students, educators, and parents have a more effective hands-on learning tool that enhances engagement and conceptual understanding in STEM environments.

2. Project Objectives and Requirements

2.1 Design Objectives

A successful project will improve the visual representation of real-world rollercoasters in K’Nex models to increase their educational value. Through a motor-controlled cart, the model will display evenly scaled kinematics.

2.2 Design Requirements

The design centers on a filament-driven extruder wheel assembly that reliably interfaces with the track and controls the cart’s motion while remaining practical to manufacture. Reliability can be evaluated by testing across multiple track layouts and confirming consistent kinematics under identical inputs. Structural components, including the seat shell and battery holster, must be optimized for 3D printing, strength, wire management, and balanced weight distribution. The control system must define ideal kinematics, calculate the necessary motor controls, and communicate these controls to the cart.

3. Design Concepts Considered

3.1 Driving Mechanisms

Initial design concept for the driving mechanism revolved around the idea of having a two-part assembly, cart and track add-on. However, these designs were complex, requiring that the track add-on be designed with the track layout in mind and limiting the modularity of the set. To abstract the mode, mechanisms that could be fully contained within the cart were explored. Initial testing of a 3D printer’s filament extrusion mechanism revealed that the idea held merit for moving the cart along the track. Functionally, this concept is reversed: the rail (in this case would be the filament) is fixed, and the train (containing the “extruder”) moves. By driving the train’s previously passive road wheels directly on the rails, this design simplifies the mechanism while restoring modularity to the K’Nex ecosystem. The designs of the mechanisms are summarized in Figure 2 below.

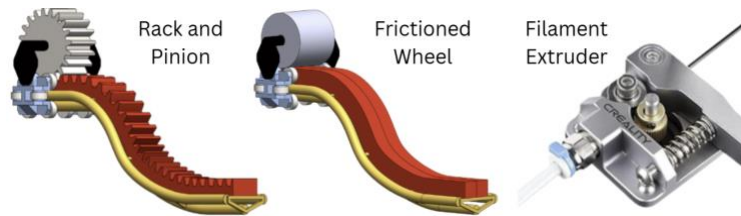


Figure 2: Driving Mechanisms (Left - Rack and Pinion Model, Middle - Friction Wheel Model, Right - Filament Extruder Concept)

3.2 Wheel Assemblies

Inspired by real roller coasters, the cart add-on includes 4 groupings of a road, up-stop, and guide wheel. While standard K'Nex cars lack guide wheels, testing showed they were necessary to limit lateral motion and provide enough normal force for the driven wheel. Initial tests revealed slipping between the driven wheel and track, indicating more friction was needed. The final design instead uses a 3D-printed wheelbase surrounded by O-rings instead of a 3D Printed toothed wheel to improve compliance and surface resolution.

4. Recommended Design Concept

4.1 Design Description

The recommended design (See Figure 3 below) uses a motor-driven filament extruder wheel assembly integrated into the coaster cart, where a rubber drive wheel presses against the track rail to actively control the cart's speed as it travels along the track. The system uses a small DC motor, a gear train, 3D-printed wheel components, and bearings to create a compact wheel cluster that both guides the cart and provides controlled propulsion. Using MATLAB and mathematically defined desired kinematics of this cart, a series of timed PWM pulses was created for the current testing track. With a web-based application, the PWM pulses are communicated to the on-board microcontroller, which drives the motor and cart.

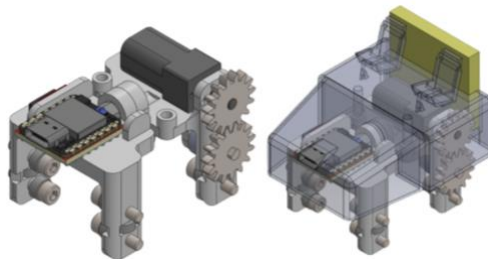


Figure 3: Final Cart Design (Left - Body Only, Right - With Shell)

4.2 Technical Basis (Analysis and/or Testing)

After mapping the testing track, the desired kinematics were calculated and visualized in MATLAB. The cart was then run on this track and recorded. Using TRACKER software, the kinematics of the base cart and the motor-controlled cart will be compared and measured for success of the project.

4.3 Key Advantages of the Recommended Design

This cart model produces sufficient contact between the wheel assembly and the track to ensure smooth motion. This proper contact also aids in ability to directly relate the motor's action to the cart's motion. The chosen motor provides sufficient torque to drive the cart while fitting within the small cart constraints. The central rotational freedom about the direction of travel allows the cart to run without interference from the track.

5. Results and Current Status

Beyond the cart prototype summarized in Section 4, tracks with increased supports of varying layout complexity were developed. The cart has been tested on both 2D tracks, with TRACKER software analysis performed on the XZ track (See Figure 4) to prove feasibility of this system to convey evenly scaled kinematics by comparing the cart's motion to the calculated desired kinematics. Initial reflection of this testing showed potential for consistent motion on drops and curves but did not display perfect scaling of kinematics.



Figure 4: Testing of Cart Model

6. Resource & Feasibility Considerations

The project's small scale and use of inexpensive materials, such as a low-power motor and 3D-printed components, keep overall costs minimal. A primary feasibility consideration is the performance and precision of 3D-printed parts due to manufacturing tolerances. Additionally, the project's specialized focus produced a knowledge constraint seen through limited local subject matter experts.

7. Recommended Improvements and Next Steps

The current cart demonstrates feasibility for integration with the K'Nex STEM Exploration Set, providing direct application for intended users. Using a different motor than originally selected has caused inconsistent torque control and runs at a slower speed than intended requiring further iterations to achieve uniform kinematic scaling. Adding a closed-loop control system would allow the model to adapt to other track layouts. Once functionality is improved, the user experience can be refined, potentially including an experiment guide outlining cart assembly, track integration, supporting processes, and the scientific purpose of the experiment.

8. Resources

- [1] "Laboratory Learning: Hands-on versus Simulated Experiments."
- [2] "Amazon.com: K'NEX Education - STEM Explorations: Roller Coaster Set , 546 Pieces, Creative Building Set, Ages 8+ : Toys & Games," Amazon. Accessed: May 13, 2025. [Online]. Available: https://www.amazon.com/KNEX-Education-Explorations-Building-Construction/dp/B01B55C2KE/ref=asc_df_B01B55C2KE?mcid=9c51f89399223bb6898f74b324c173ba&tag=hyprod-20&linkCode=df0&hvadid=693442482467&hvpos=&hvnetw=g&hvrnd=14255832334557443454&hvpone=&hvptwo=&hvqmt=&hvdev=c&hvdvcmld=&hvlocint=&hvlocphy=9001982&hvtargid=pla-348682312010&psc=1&hvocijid=14255832334557443454-B01B55C2KE-&hvexpln=0
- [3] Pincus, Jared, "Properly Scaling the Speed of a Model Roller Coaster," presented at the Bridges 2024 Conference, Aug. 2024. [Online]. Available: <https://archive.bridgesmathart.org/2024/bridges2024-495.pdf>